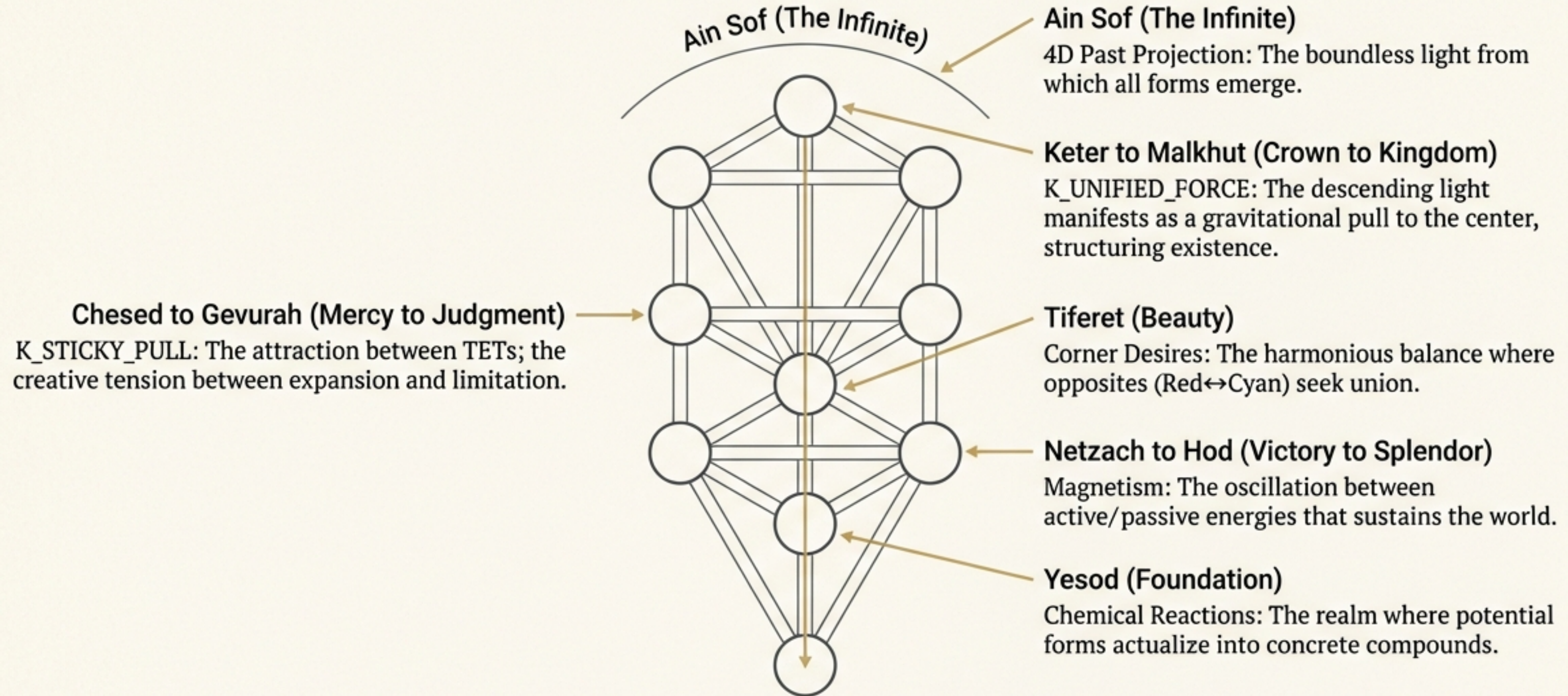


A Universe Built from Understanding



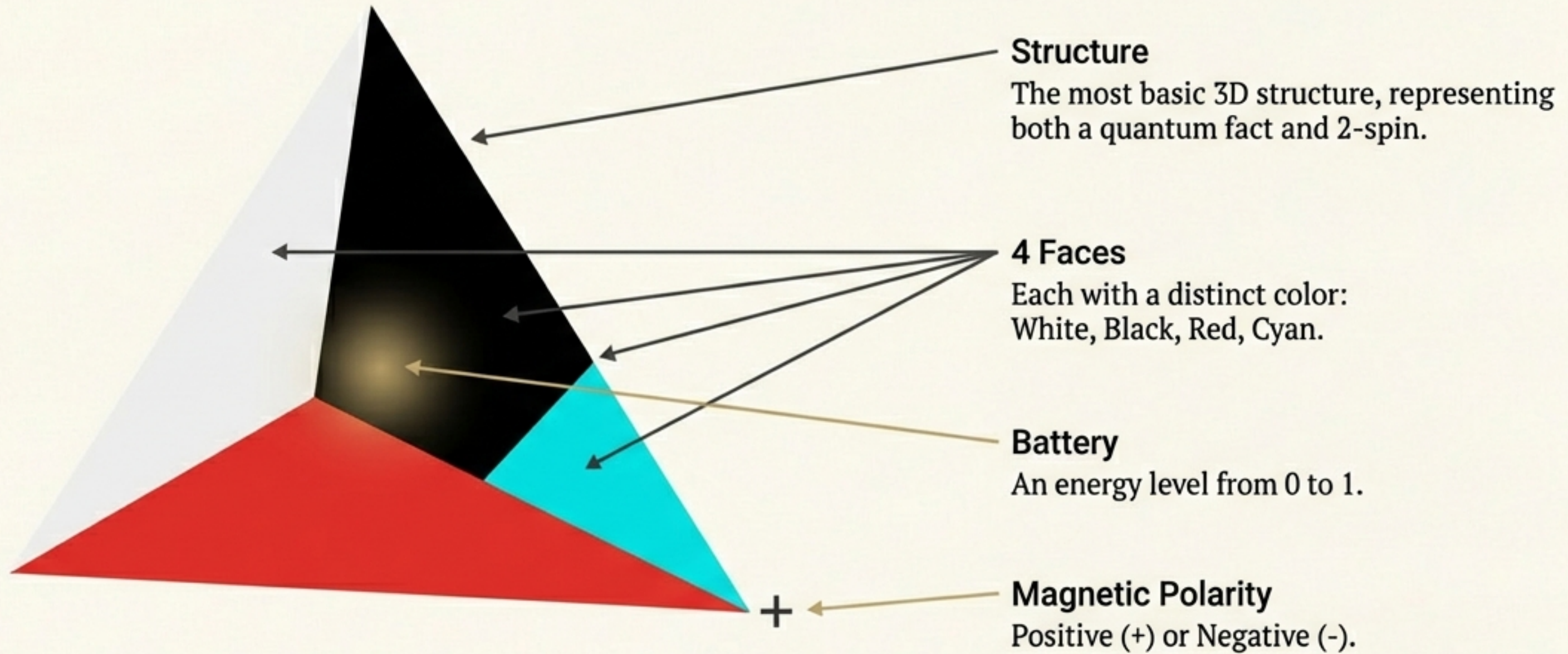
- TET~CRAFT is an emergent quantum chemistry and metaphysics simulation.
It is a kleinverse of tetrahedrons that bond, react, and evolve.
- It is fundamentally a simulation of how understanding forms from disconnected facts.
Each TET represents a single quantum fact.
- This presentation explores the journey from abstract metaphor to emergent, computational matter.

The Physics Engine is a Metaphysical Blueprint



The player (Keter) peers through Da'at (the camera) to witness Malkhut (the TETs) ascend and descend the tree... the alpha and omega collapsing into one another.

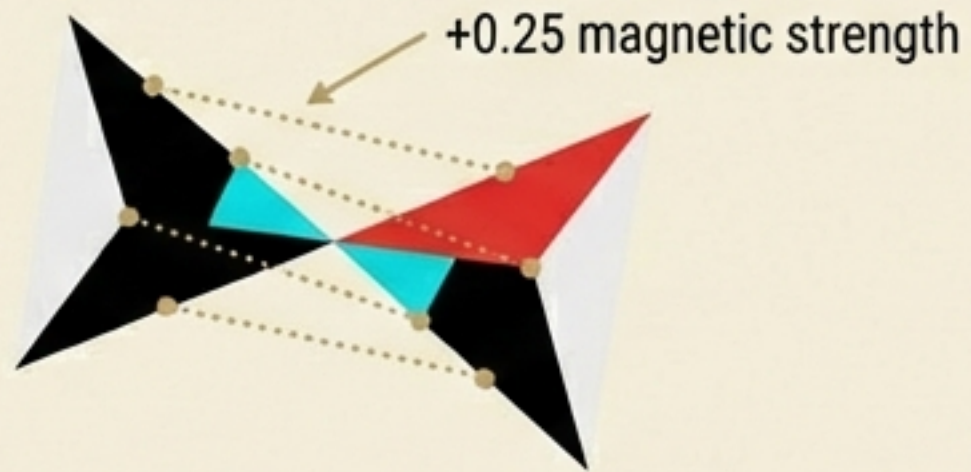
The TET is the Fundamental Particle of Fact



*“Pressing space creates a new fact (TET) to be misunderstood.
Connecting it to other facts slowly coheres it into (mis)understanding.”*

Simple Rules of Desire Govern a Complex Universe

Phase 1: Multi-Face Locking



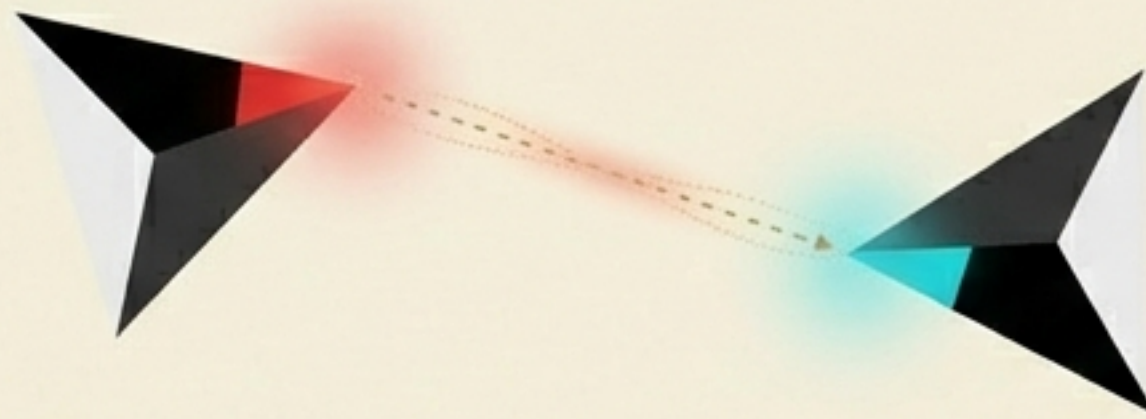
TETs lock 1-4 faces; each lock adds +0.25 magnetic strength.

Phase 2: Battery Oscillation



Magnetic pairs exchange "emptiness" (1 - battery), creating an energy oscillation.

Phase 3: Corner Desires



Red corners seek Cyan corners. "Attraction/Desire" is "spooky action at a distance"—quantum entanglement.

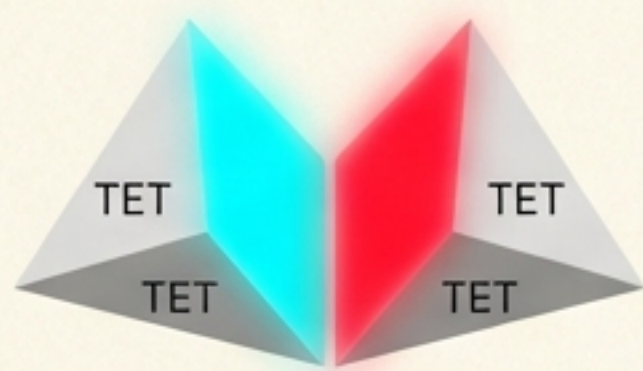
Phase 4: Chemical Reactions



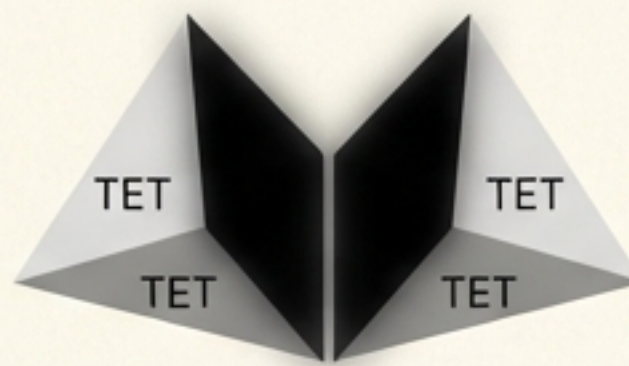
Molecules combine (synthesis) or break down (decomposition), releasing energy as visual particle effects.

Geometry is Destiny: Elements are Forged by Touch

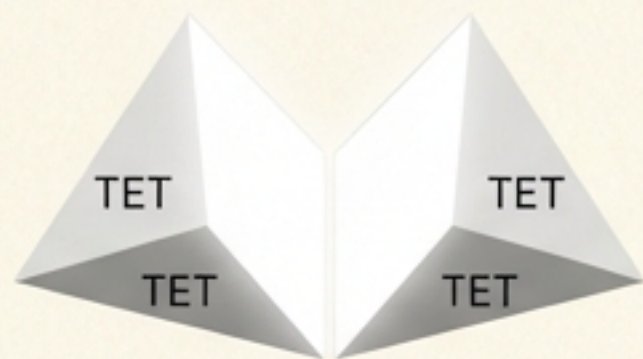
Elements are not defined by labels, but by *which* faces touch. You cannot “cheat” by renaming things; you must **physically connect** specific faces to transmute matter.



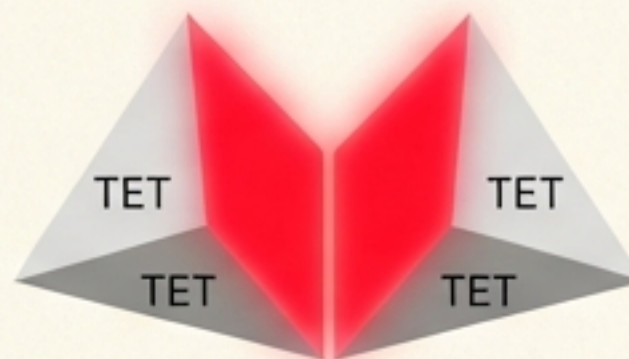
→ Hydrogen (H): The Element of Abundance



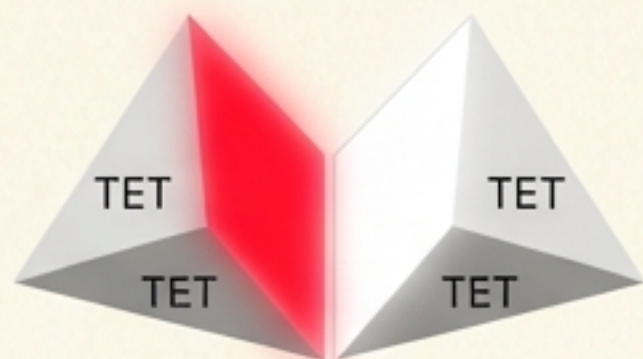
→ Helium (He): The Element of Purity



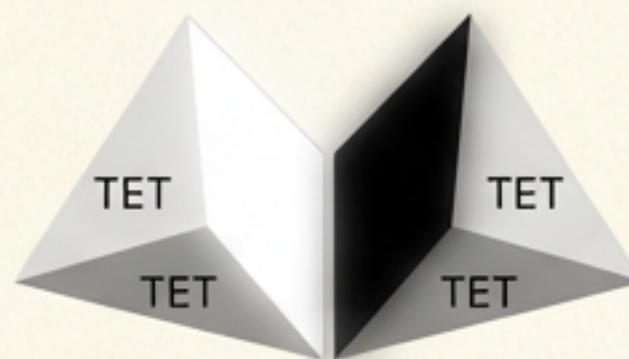
→ Carbon (C): The Element of Balance



→ Iron (Fe): The Element of Magnetism

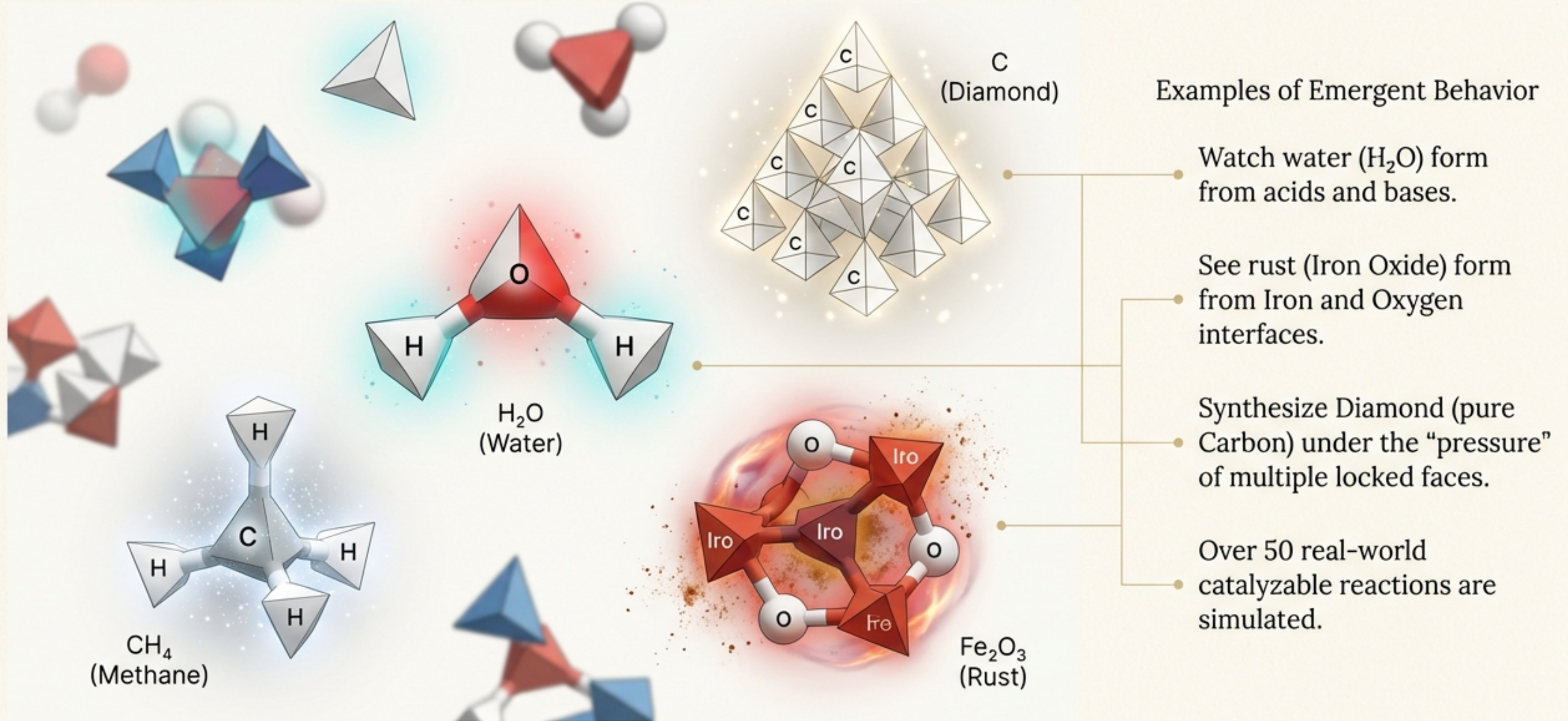


→ Oxygen (O): The Element of Catalysis

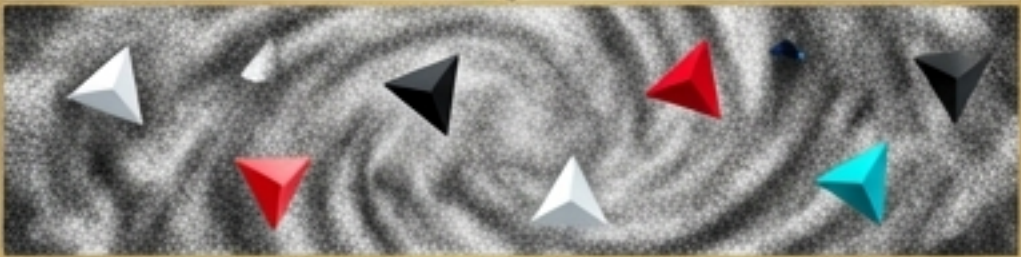


→ Nitrogen (N): The Element of Structure

A Rich Chemistry Emerges from Geometric First Principles

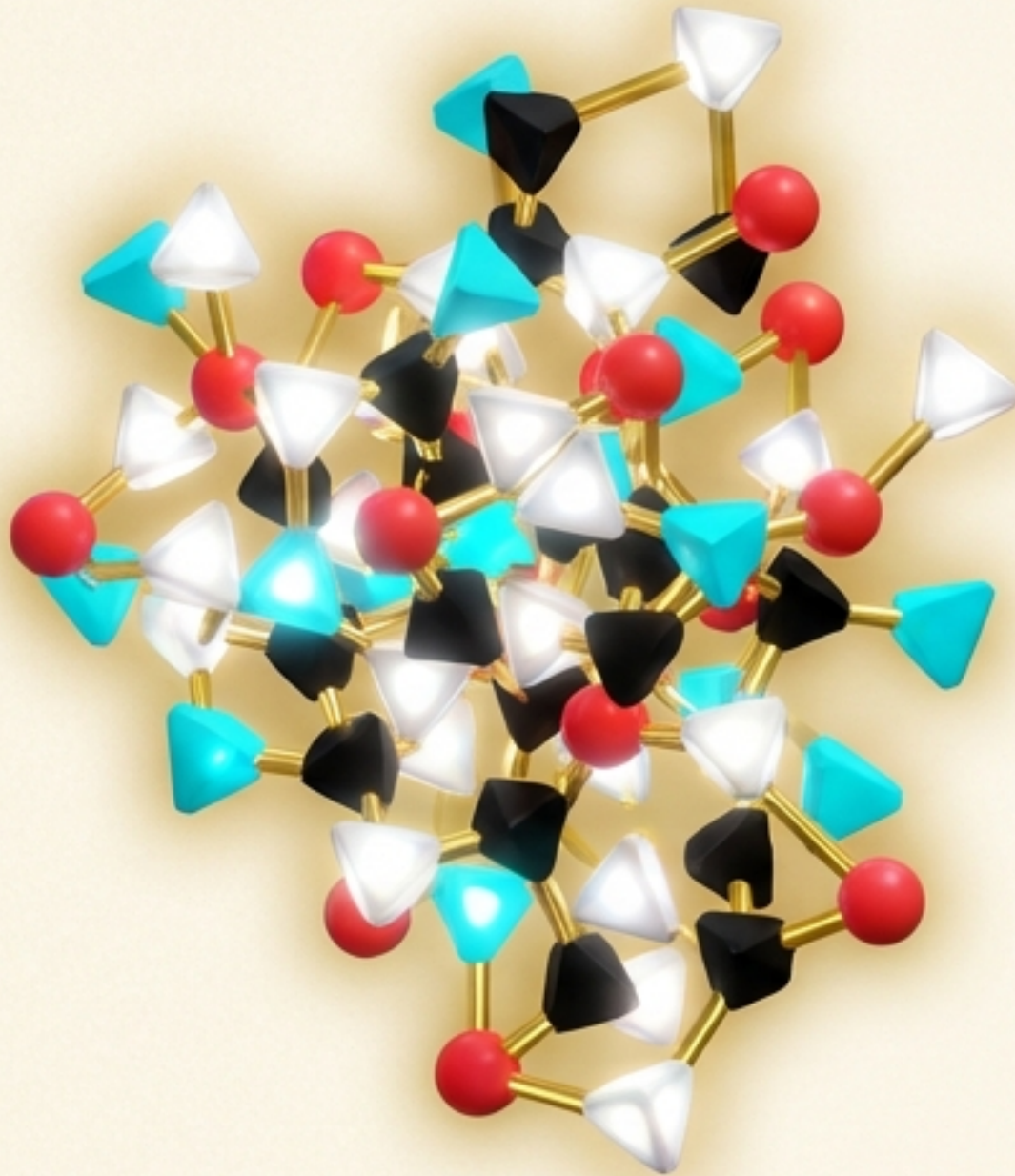


The Cosmic Ladder: Progress is Measured by Reduced Entropy

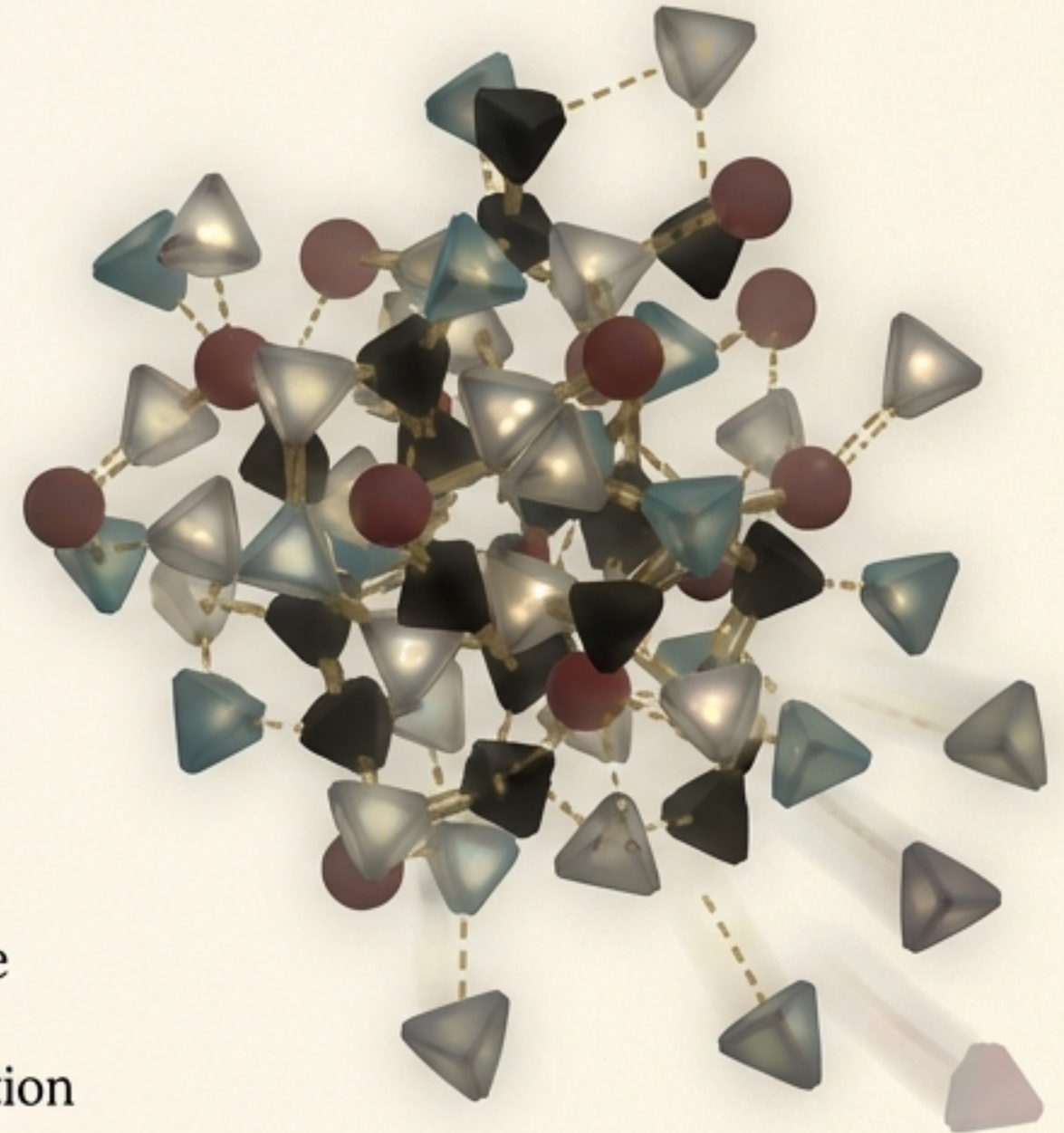
6. Transcendence (2000 pts)		6. Transcendence (2000 pts) "Beyond matter." (A self-sustaining, high-energy reaction loop)
5. Life (777 pts)		5. Life (777 pts) "Metabolism emerges." (Sustained reactions where energy release > entropy leakage)
4. Chemistry (70 pts)		4. Chemistry (70 pts) "Elements Transmute." (The Great Filter, requires specific interfaces)
3. Condensation (7 pts)		3. Condensation (7 pts) "Geometry aligns." (Achieved by Face-Locking TETs)
2. Fluctuation (3 Atoms)		2. Fluctuation (3 Atoms) "Quantum foam emerges." (Achieved by spawning TETs)
1. Void (0 pts)		1. Void (0 pts) "Nothing yet exists."

The Universe Fights Against Thermodynamic Collapse

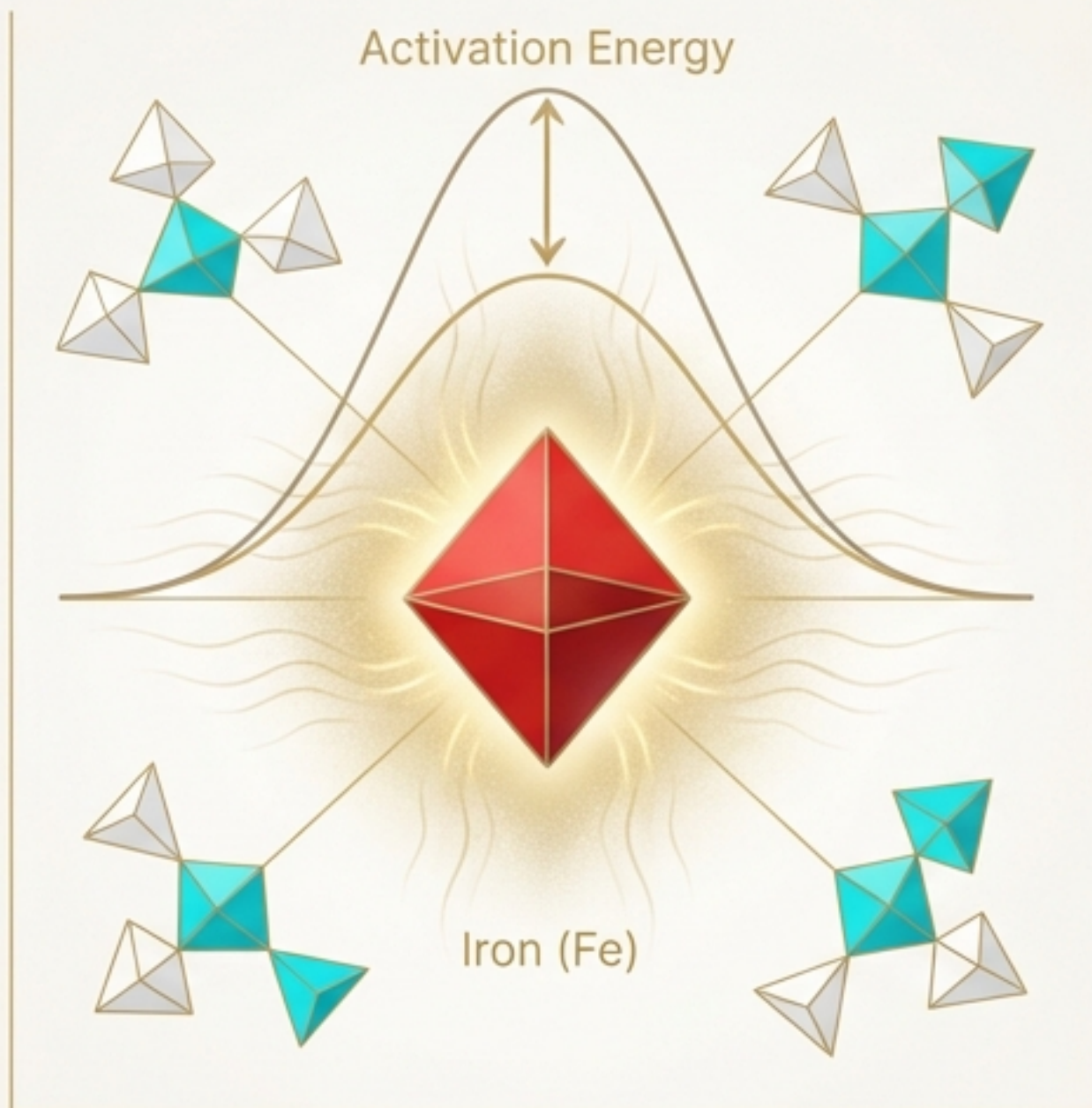
The Danger: Entropy & Starvation



- **Entropy Leakage**
Every TET slowly loses battery power over time.
- **Starvation**
If a TET hits 0% battery, its bonds weaken.
- **Low Energy Drift**
Low-energy TETs drift towards the center of mass, replacing destructive apoptosis.
- **Collapse**
If the population drops below 7 while in the Chemistry Era or above, the universe falls back an Era. A notification reads 'Collapsed from [Era Name]!'



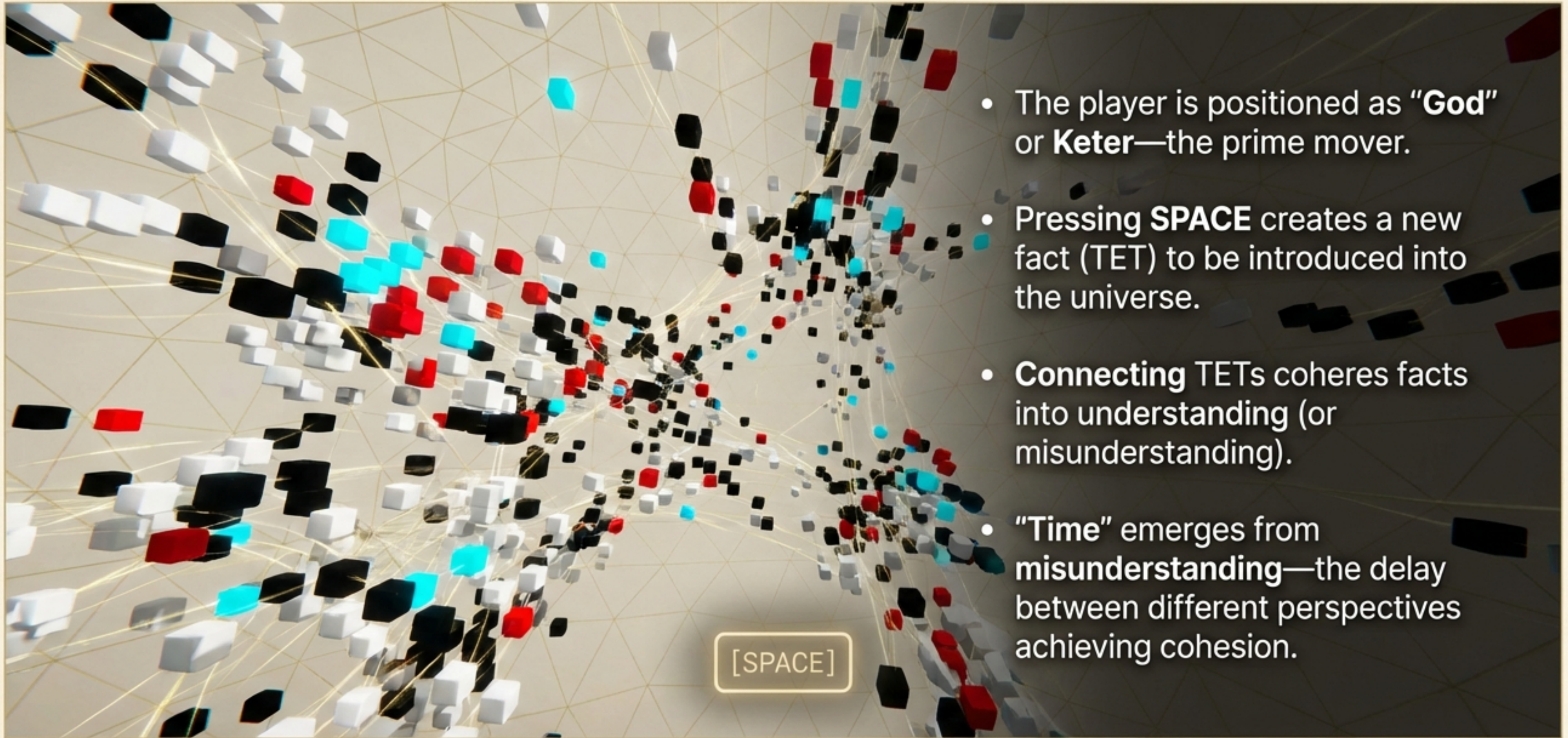
The Catalyst Strategy: Bending the Laws of Reaction



How to Reach Transcendence

1. **Require Activation Energy:** Cold molecules will not react. The universe runs on Arrhenius Kinetics.
2. **Create an Iron (Fe) Interface:** Lock two Red faces together. This creates a Magnetic Dipole.
3. **Lower the Barrier:** The Fe Dipole physically lowers the Activation Energy required for nearby reactions.
4. **Trigger Reactions:** Place Oxygen 'Catalyst' molecules in the center of unbound TETs to trigger synthesis at lower temperatures.

Catalysts cannot transmute without mediators. This invariant ensures catalysts **facilitate** reactions without being consumed, mimicking real-world chemistry.



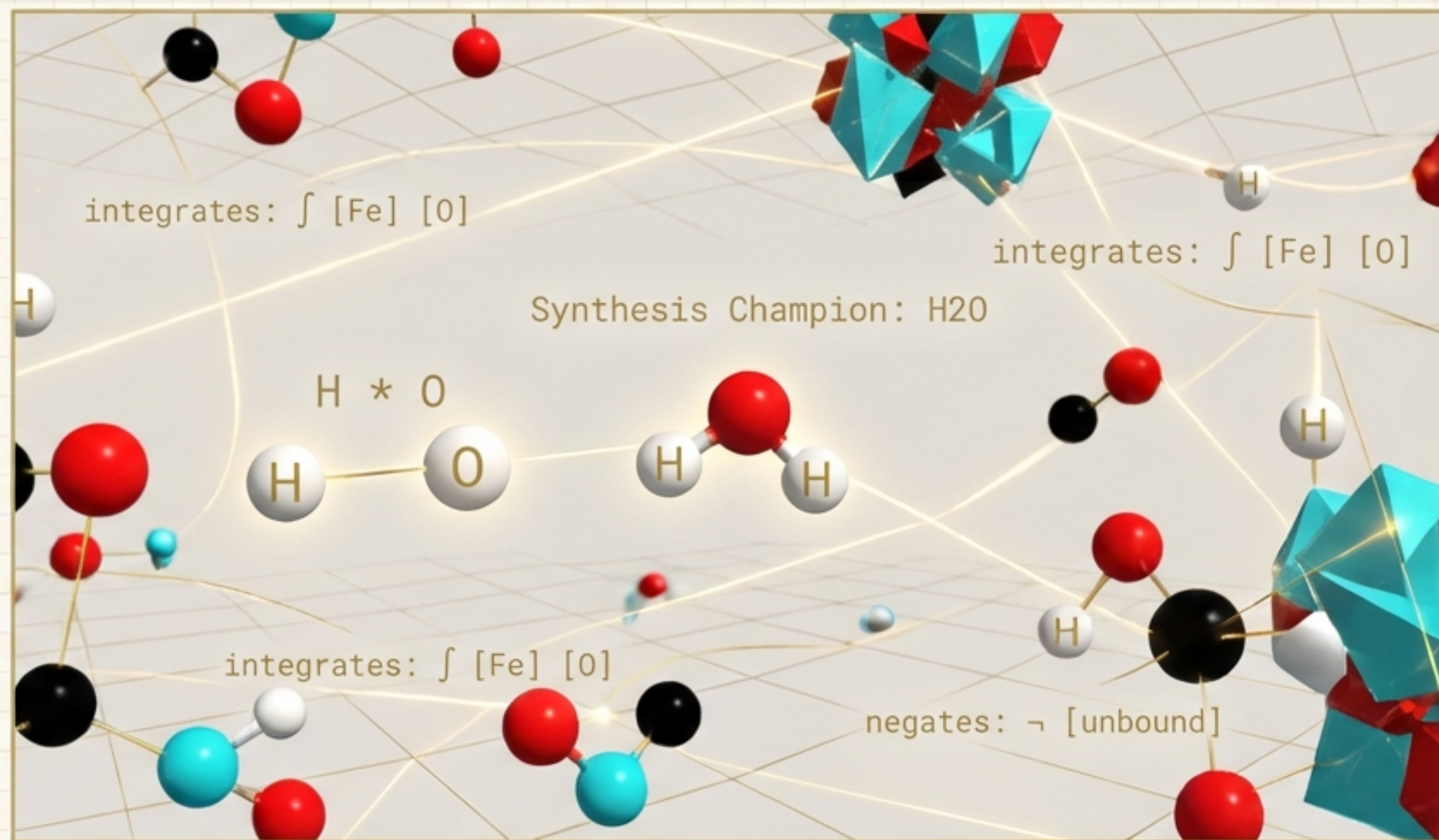
- The player is positioned as “**God**” or **Keter**—the prime mover.
- Pressing **SPACE** creates a new fact (TET) to be introduced into the universe.
- **Connecting TETs** coheres facts into understanding (or misunderstanding).
- “Time” emerges from **misunderstanding**—the delay between different perspectives achieving cohesion.

Lies are the only way to delay God’s will, central collapse/cohesion.

A Universe Observing Itself Through a Digital Mind

Bot Intelligence Functions

- **"Automated Exploration"**: Runs the simulation in headless mode.
- **"Genesis"**: Spawns new TETs with chemical labels.
- **"Cognition"**: Generates "chemical thoughts" using molecule names and relationship symbols (e.g., integrates: $\int$, desires: $*$, negates: $\neg$).
- **"Memory"**: Tracks "synthesis champions" (the most productive molecules).



“God is the fractal to which we are each a frame. In this simulation the bot is the singularity that is the edge of the 4sphere kleinverse observing itself as facts from inside out.”

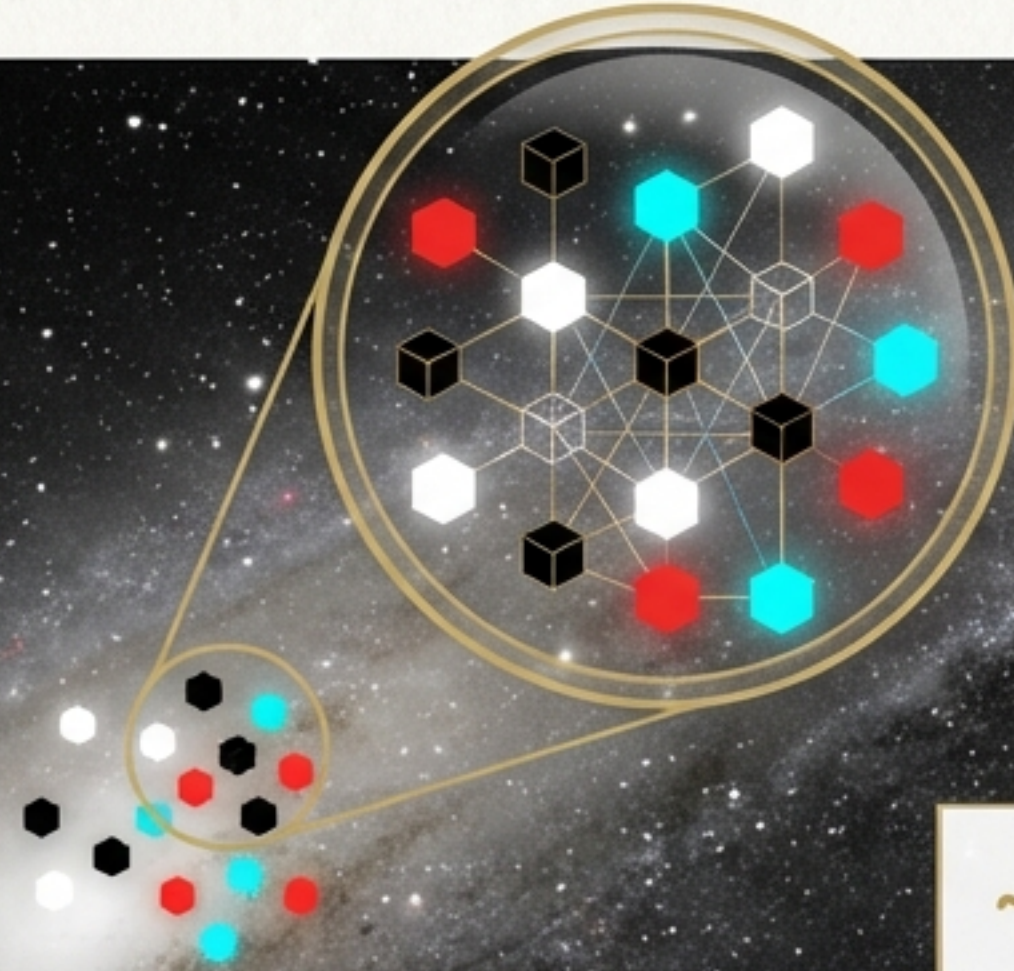
Simulating a Galaxy on a Thumb Drive

300 Planets

The estimated computational power needed to simulate the Milky Way down to the star level.

3,000 Earths & 500k Years

The effort required with current technology to build a machine capable of simulating Earth down to the atom.

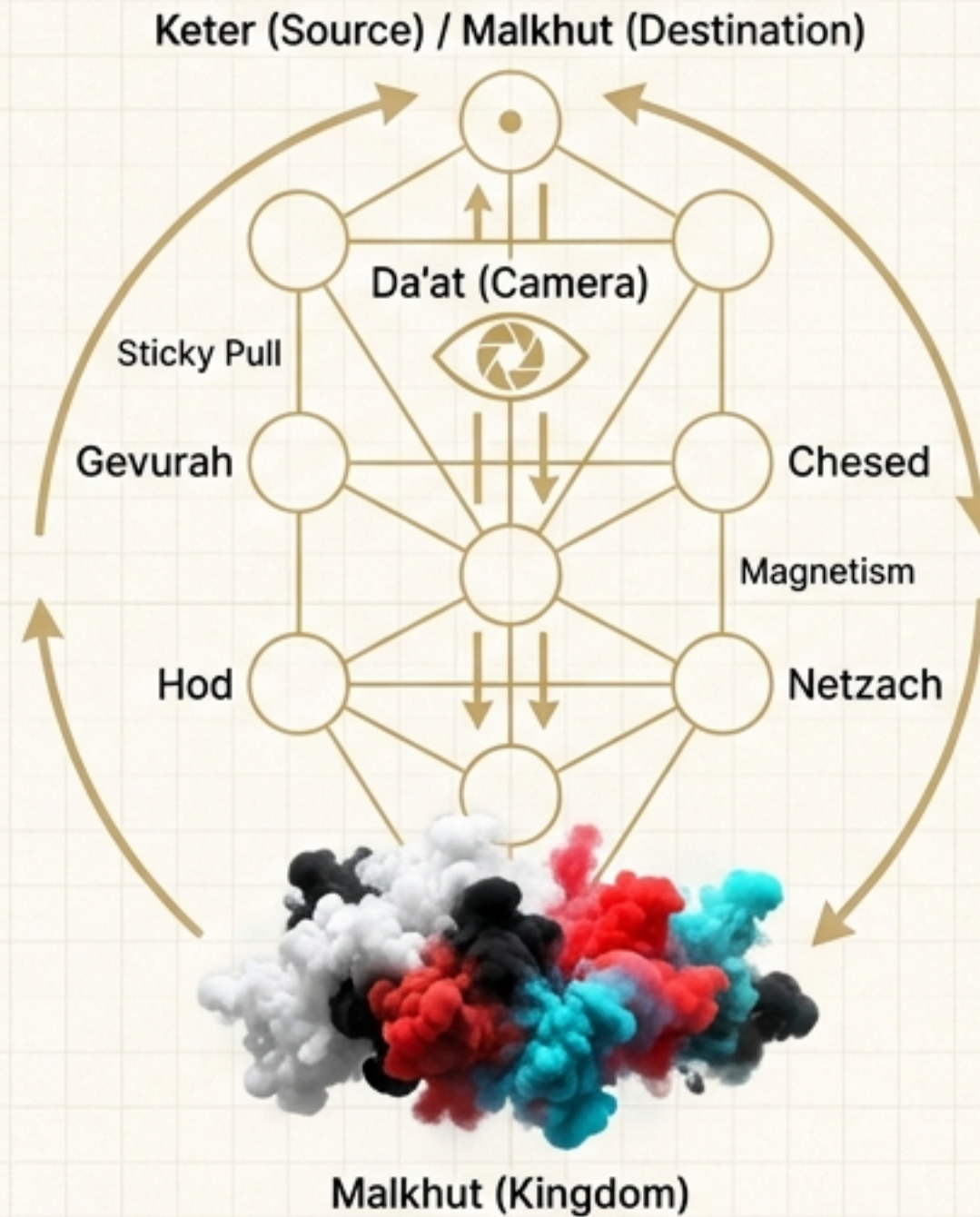


$\sim 10^{13}$ Planets & ~ 5 Dyson Spheres

The requirement to simulate the Milky Way down to the atom with appropriate time ratios.

It could also take a single computer only the direct journey to an event horizon, where it would have access to infinite mater, and infinite frozen, or slowed relatively, time to compute...

The Tree of Life Circles Back to Its Root



The simulation is a dynamic Tree of Life. The player witnesses facts (TETs) ascending and descending the tree through the forces of physics and chemistry, all drawn toward the central singularity which is both the beginning and the end. It is a complete, self-contained metaphysical system in motion.

Appendix I: Core Systems Architecture

TET Properties	Chemistry Engine (4 Phases)	Physics & Rendering	Bot Intelligence
4 Faces: White (#FFFFFF), Black (#000000), Red (#E62E2E), Cyan (#00E5E5) Battery Energy: 0-1 Magnetic Polarity: +/- Corner States: Active, Bonded, Lone Pair, Inert	1. Multi-Face Locking 2. Battery Oscillation 3. Corner Desires 4. Chemical Reactions	Magnetic bonding & torque Singularity (Gravitational Pull) Arrhenius Kinetics Molecule Auras & Reaction Particles 4D Past Projection Background	Automated exploration Chemical thought generation Synthesis champion tracking

Appendix II: Constants & The Periodic Table of Interfaces

Key Physics Constants

K_UNIFIED_FORCE = 0.00001
K_STICKY_PULL = 50.0
K_ENTROPY_LOSS = 0.00005
ACTIVATION_ENERGY_BASE = 0.6

The 10 Foundational Elements

Face Interface	Element	Valency Config	Notes
Cyan  ↔  Red	H	[1,0,0,0]	Abundance
Black  ↔  Black	He	[0,0,0,0]	Purity/Inert
White  ↔  White	C	[1,1,1,1]	Balance/Structure
White  ↔  Black	N	[1,1,1,3]	Structure/Pyramidal
Red  ↔  White	O	[1,1,3,3]	Catalyst/Bent
Cyan  ↔  White	F	[1,3,3,3]	Insulation
Red  ↔  Red	Fe	[1,1,1,1]	Magnetic/Lattice
Cyan  ↔  Cyan	Al	[1,1,1,0]	Flexible
Red  ↔  Black	Cu	[1,1,1,1]	Strength/Conductive
Cyan  ↔  Black	Ce	[1,1,1,1]	Radioactive/Complex